Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

HHS Public Access

Author manuscript

J Biol Rhythms. Author manuscript; available in PMC 2025 December 11.

Published in final edited form as:

J Biol Rhythms. 2026 February ; 41(1): 42–52. doi:10.1177/07487304251392289.

Circadian phase estimation from ambulatory wearables with particle filtering: accuracy depends on initialization, recording duration, and light exposure

Lara Weed[1], **Arec Jamgochian**[2], **Melissa A. St. Hilaire**[3], **Philip Cheng**[4], **Mykel J. Kochenderfer**[2], **Jamie M. Zeitzer**[5,6]

[1]Department of Bioengineering, Schools of Engineering & Medicine, Stanford University, Stanford, CA 94305

[2]Department of Aeronautics and Astronautics, School of Engineering, Stanford University, Stanford, CA 94305

[3]Department of Computer and Data Science, School of Engineering & Computational Science, Merrimack College, North Andover, MA 01845

[4]Henry Ford Health System, Detroit, MI 48202

[5]Department of Psychiatry and Behavioral Sciences, School of Medicine, Stanford University, Stanford, CA 94305

[6]Mental Illness Research Education and Clinical Center, VA Palo Alto Health Care System, Palo Alto, CA 94304

Abstract

While current mathematical models of human circadian rhythms accurately predict circadian phase responses to light in controlled laboratory experiments, they show reduced performance in the real world, especially among shift workers with irregular schedules and downstream erratic light diets. The source of the discrepancy between in-laboratory and ambulatory performance remains unclear. We evaluate the impact of initialization strategy, recording duration, and light exposure characteristics on model performance using wearable data from both individuals on regular schedules and shift workers. We implement a probabilistic initialization framework to account for unknown starting phase and assess model performance in prediction of phase from light input data against an in-lab measure of circadian phase (dim light melatonin onset). In

laraweed@stanford.edu .

Author Contributions: Conceptualization: L.W. and J.Z.; Methodology: L.W. and A.J.; Software: L.W.; Formal analysis: L.W., A.J., and J.Z.; Investigation: L.W., A.J., J.Z.; Resources: J.Z.; Data Curation: L.W., J.Z., and P.C.; Writing - Original Draft: L.W.; Writing - Review and Editing: L.W., A.J., M.S., P.C., M.K., and J.Z.; Visualization: L.W.; Supervision: J.Z., Project administration: L.W.; Funding Acquisition: L.W. and J.Z.

Ethical Considerations: Data supporting this research was previously collected, and approved under Stanford IRB-10165 and Henry Ford Health IRB-11468 and IRB-8662.

Consent to Participate: Not applicable

Consent for Publication: Not applicable

Declaration of conflicting interest: The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

participants with regular schedules, accuracy improved with longer recordings, while shift workers show no accuracy gains when having more nights of data. Light exposure patterns differed significantly between groups, with brighter and more regular day-to-day light exposure being weakly to moderately associated with improved model estimates, whereas fragmented patterns of light exposure increased uncertainty. These findings suggest that current models require adaptation, particularly in light sensitivity, to generalize to free-living, irregular conditions and support robust, scalable circadian tracking in real-world populations.

Keywords

Mathematical models; Circadian Rhythms; Wearable Devices; Probabilistic models; Shift Work Schedule; Light

Introduction

Accurate determination of the timing of the circadian clock (phase) outside of controlled laboratory environments is a key milestone for the field of circadian rhythms. Indirect markers of circadian phase, such as dim-light melatonin onset (DLMO) or core body temperature minimum (CBT_{min}), are commonly used, though these are only obtainable under intensive, highly controlled protocols that account for physiology and behavior that can mask the underlying circadian components (Duffy and Dijk, 2002). Recent advances in wearable sensing technologies have enabled remote, longitudinal measurement of light exposure, the primary zeitgeber that entrains the human circadian central clock (Duffy and Czeisler, 2009). Using light data, in theory, one can estimate the position of the circadian clock independent of masking variables, such as activity, posture, and food intake (Mrosovsky, 1999). While this presents opportunities for measurement of the circadian clock in the real world, it also introduces new challenges: unknown initial states for models, variable recording durations, and discrepancies between wrist-worn and ocular light exposure.

Prior research on estimating circadian phase using data from wearable sensors has spanned multiple modalities, including actigraphy as a proxy for behavior (e.g., sleep onset, mid-sleep timing) and physiological measurements (e.g., heart rate, skin temperature, breathing rate) (Huang et al., 2021). Although the daily pattern of many behavioral markers is driven by the circadian clock, there are common behaviors that can obscure the underlying circadian component (e.g., exercise, self-selected erratic sleep timing, eating), making these inadequate markers of the phase of the central circadian clock under most ambulatory conditions (Huang et al., 2021, Mrosovsky, 1999). Even employing statistical techniques, such as machine learning, to tease these relationships apart has not yielded a reliable tool for estimating circadian phase (Brown et al., 2021).

A promising avenue for estimating circadian phase, developed using in-laboratory experimental data, couples patterns of light exposure with mathematical models of the human circadian clock. Existing models transform ocular light exposure into a circadian drive that regulates the dynamics of a modified van der Pol oscillator representing the central clock (Kronauer et al., 1999, Forger et al., 1999, St. Hilaire et al., 2007). These

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

models perform well in laboratory studies with controlled light exposure (low minute-to-minute variability in intensity and spectral composition, square-wave patterns), but their performance degrades in ambulatory settings (Huang et al., 2021). Poor performance could be due to initialization strategies, which vary widely, often relying on steady-state assumptions or synthetic light histories (Huang et al., 2021, St.Hilaire et al., 2007, Woelders et al., 2017, Stone et al., 2019, Mayer et al., 2025). While such strategies function under regular schedules, they fail to accommodate the variability in shift workers or those with erratic schedules (Stone et al., 2019).

Estimation theory approaches, including Kalman and particle filters, are a promising method to estimate circadian phase by combining wearable-derived inputs with dynamic models (Bonarius et al., 2021, Mott et al., 2011, Kim et al. 2023a & 2023b). These methods enable probabilistic tracking and uncertainty quantification and have been extended to support personalized model parameters and input noise modeling. However, most applications focus on physiological outputs rather than light-driven inputs, and few address practical challenges such as initialization and variable data duration, which are key focus areas of this study.

Given these gaps, our central question is whether the decrease in performance of circadian models evaluated on wearable data arises from model limitations or from mismatches between model assumptions and real-world conditions. We begin by evaluating the role of initialization using a probabilistic framework to quantify its impact and reduce bias. We then examine how recording duration affects phase estimation accuracy and precision across schedule types. Finally, we explore how light exposure pattern characteristics influence model performance. This work aims to provide practical insights for improving circadian phase estimation from wearable light data under real-world conditions.

Methods

Datasets & Preprocessing

Data sets derived from individuals with regular schedules and those participating in shift work were used to examine circadian model performance with ambulatory light data from wrist-worn wearables. The regular schedule dataset contained 56 healthy participants (age: 27.1 ± 4.6 years; 59% male; 53.6% white, 26.8% Asian, 7.2% Black, 12.5 % multiple races/other/not reported; 12.5% Hispanic or Latino), who were instructed to go to bed and wake up at consistent times each day, keeping both within a one-hour window, and ensuring that the time between bed and wake times were within 6-9 hours. Participants were instrumented with a wrist-worn actigraph (Motionlogger, Ambulatory Monitoring Inc, Ardsley, NY) that measured illuminance (range: 0 – 2,700 lux , resolution: 1 lux, 30s epochs) for two weeks before an in-laboratory dim light melatonin onset (DLMO) assessment under constant routine conditions that determined the criterion standard for the phase of the central circadian clock (Duffy and Dijk, 2002). Data were previously collected by the Zeitzer Circadian Research Lab at Stanford University under IRB-10165 and published in Najjar and Zeitzer, (2016) and Joyce et al. (2022a and b).

The shiftwork dataset contains 32 shift workers (age: 37.5 ± 11.3 years, 15.6% male, 53.1% Black, 31.2% white, 15.6% multiple races/other/not reported; 0.0% Hispanic or Latino),

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

who were engaged in a variety of shift work schedules (e.g., night work, rotating shifts) over two weeks prior to in-laboratory DLMO assessment. Shift workers wore a wrist-worn actigraph (Actiwatch 2, Philips, Murrysville, PA) that measured illuminance (range: 5 - 100,000 lux, 30s epochs) during these two weeks. Data were collected by the Cheng Lab at Henry Ford Health under IRB-11468 and IRB-8662 and previously published in Cheng et al. (2021).

The two datasets used different devices in the recording of light exposure patterns. However, both were recorded from the wrist at the same sampling rate and within a physiologically relevant range for circadian phase impact (Zeitzer et al., 2000). The range of the device used in the shift worker population was larger but both span from dark to sufficient illuminance to distinguish between dim, indoor, and outdoor lighting.

Data preprocessing was completed using MATLAB (MathWorks Inc., Natick, MA). Data were screened for missing timestamps and nonwear. Files containing fewer than three days of data or with a missing data ratio of greater than one day per week were excluded from analysis. Median imputation was performed on missing data bouts of greater than 1 hour impacting only one file (Weed et al., 2022). Additionally, since the regular schedule dataset had a maximum value of 2,700 lux, we artificially bound the shiftwork dataset to match, though this was not anticipated to change the outcome of the models given that this illuminance produces a near saturating circadian response (Zeitzer et al., 2000).

Models

Light data were used as inputs into a previously developed model of the human central circadian clock (St. Hilaire et al., 2007). This model considers illuminance and timing of light exposure to estimate circadian drive (Process L) that regulates a modified van der Pol oscillator representing the central clock (Process P) to predict circadian phase markers (e.g., DLMO, CBT_{min}). We examined a modified version (St. Hilaire et al., 2007) of the Jewett-Forger-Kronauer model (Kronauer et al., 1999, Forger et al., 1999) accounting for light sensitivity at illuminances < 150 lux with the following form for light sensitivity in Process L:

$$\alpha(I(t)) = \alpha_o(\frac{I(t)}{9500})p(\frac{I(t)}{I(t)+100})$$

where α_o is 0.1 min^{-1}, $I(t)$ is light illuminance (lux), and p is 0.5. All parameters for the model were set and estimates of phase marker timing were extracted according to standard procedures (St. Hilaire et al., 2007). Ordinary differential equation (ODE) solvers (ODE15s in Matlab) were used to solve for model solutions accounting for the time-varying input of the light exposure data. CBT_{min} timing estimates minus 6.67 hours were used to estimate DLMO timing, consistent with previous literature (St. Hilaire et al., 2007 (*J Pineal Res)*, Burgess et al., 2003). The St. Hilaire model was fit previously based on in-laboratory phase marker assessments (DLMO, CBT_{min}) and light exposure experimentally delivered. Models have been validated for light exposure 0.05-10,000 lux.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Model Initialization and Particle Filtering

ODE solutions are sensitive to their initializations. In real-world applications, however, the initial state is often unknown. To address this, we used a probabilistic approach to examine the impact of initial conditions on the final DLMO timing estimate. Absent any initial information, we assumed a uniform prior in which we approximated using 1440 state particles evenly distributed across phases (one per minute of the day). The initial state of light sensitivity in the prompt-response portion of process L was initialized at 0.5 as in previous work (St. Hilaire et al., 2007), and amplitude was initialized at 1.0 across all models since the dynamics of both are much faster than phase dynamics as evident by model equations and have little impact on the final estimated phase. We propagate the particles through the model to estimate a posterior containing estimated DLMO timing error for each light diet recorded from wrist-worn sensors.

Model performance was characterized over participant DLMO timing distributions produced after running models with multiple initializations. We used a Gaussian kernel density estimate with bandwidth determined automatically to estimate the maximum likelihood estimate (MLE) of DLMO timing and negative log likelihood (NLL) of the correct timing under the predicted distribution. A von Mises distribution was also fit to the data to capture circular distribution center (μ) and concentration (K). This was used to compute the circular entropy to characterize the spread of predictions, with larger values indicating a more uniform and less concentrated distribution with more flexibility. To assess performance over each dataset, the mean root-mean-square error (RMSE) of participant MLEs of DLMO timing was computed. We also computed the median and IQR of each performance metric.

Light Diet Features

Model performance metrics were compared against a variety of features from individual light diets. These included illuminance-based features (mean, median, 95th percentile), temporal features (percentage of samples >0 lux or >1000 lux, interdaily stability, intradaily variability), and duration of data collection (Weed et al., 2022, Witting et al., 1990). To jointly capture temporal and intensity structure, we also performed functional principal component analysis (fPCA), which applies Fourier smoothing followed by functional component analysis across days (Zeitzer et al., 2013). fPCA was conducted on the 14-day light data prior to the in-lab phase assessment. Data were aligned to start at 00:00 prior to smoothing. Each orthogonal component (eigenvector) derived from the functional data analysis was associated with a score (eigenvalue) that was used in statistical testing.

Statistical Analysis

We first compared the two datasets to examine differences in light diet features. After testing for normality, we used a Mann-Whitney U test to compare illuminance-based features, temporal features, duration of data collection, and the first four component scores from fPCA.

We then extracted performance metrics from the particle filtering approach to circadian modeling and examined the influence of recording duration and number of initializations on performance metrics. To examine the impact of the recording duration on model

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

performance, we started models on one-day increments from DLMO timing up to the total recording length using 1440 initializations. Model performance metrics were extracted for each initialization and recording length scheme. Lin's concordance was also computed for select models.

To examine the impact of the number of initializations on model performance, we selected random permutations of particles from the 1440 initialization particles and computed the dataset mean over the performance variables. Each number of initializations was repeated 100 times and the standard error was computed for each performance metric. We assessed initializations ranging from 1-1440 over a two week period preceding in-lab DLMO assessments. Lin's concordance was also computed for select models.

Finally, we examined the relationship between performance metrics and light diet features. Cross-correlations (Pearson correlation coefficients) were computed between light features and model performance (mean RMSE and entropy) to assess the influence of light diet on outcomes. Data are presented as median [IQR] unless otherwise noted.

Results

Dataset Characteristics

Light recording characteristics are reported in Table 1. Recording length was significantly longer in the shift workers compared to the regular schedule dataset, but exceeded 14 days for most participants (77% and 75%, respectively). The participants in the regular schedule dataset recorded a median of 20% more time in non-zero light exposure compared to the shift work dataset. Similarly, the mean and median lux was greater in the regular schedule dataset by 54.0 lux and 14.6 lux, respectively. Interdaily stability of light patterns was significantly lower in shift workers indicating less regular day-to-day light exposure compared to the regular schedule dataset. There were no significant differences between regular schedule dataset and shift work datasets in proportion of time spent above 1000 lux, the 95th percentile illuminance, and consolidation of light exposure (intradaily variability).

Functional principal component analysis enables one to visualize common patterns in 24-hour light diets (Figure 1). Qualitatively, principal component (PC) 1 represents the overall daylight intensity (variance explained: 58.1%). PC2 represents the relative contribution of elevated light exposure in either the beginning or end of the day (variance explained: 17.6%). PC3 represents the duration of light exposure (variance explained: 13.4%). PC4 is more complex but generally represents a morning or evening timing shift in light exposure, likely associated with the timing of nocturnal sleep (variance explained: 6.8%). Comparing the datasets (Table 2), regular schedule participants had significantly higher overall daylight intensity (PC1) and an earlier peak of light exposure (PC2). There were no detectable differences between datasets in duration of light exposure (PC3) or light timing shift (PC4).

Recording Duration and Model Performance

Recording duration affected all model performance metrics (Figure 2). Precision improved with longer recordings, reflected by lower entropy and higher K, and did so more rapidly in the regular schedule dataset than in the shift work dataset. In the regular schedule dataset,

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

accuracy also improved with increased recording length, indicated by lower NLL & RMSE values and MLE & μ estimates closer to zero. In contrast, the shift work dataset showed gains in precision (reduced entropy) with longer recordings but did not exhibit consistent improvements in accuracy (RMSE). Lin's concordance for a recording duration of 14 days with all 1440 initializations was 0.93 for the regular schedule dataset and 0.89 for shift workers.

Initializations and Model Performance

Initialization count influenced model performance for 14-day recordings (Figure 3; see SI 1 for all 1440 initializations). With too few initializations, both precision and accuracy are mischaracterized (entropy is artificially low, K is inflated, and NLL is elevated). In the regular schedule dataset, entropy stabilizes (value >95% of maximum at 1440 initializations) at ~180 initializations, while the shift work dataset requires ~360 initializations. RMSE, by contrast, stabilizes with relatively few initializations, and MLE converges after ~48 initializations. Beyond ~48 initializations, recording duration has a larger impact on model performance than further increasing initialization count. Lin's concordance for a recording duration of 14 days with 48 initializations was 0.94 for the regular schedule dataset and 0.88 for shift workers.

Light Diet and Model Performance

A variety of aspects of the light diet contributed to improved model performance (Figure 4). In general, more exposure to higher illuminances improved model performance. The accuracy of the model prediction, represented by RMSE, was most robustly influenced by overall increased exposure to light. Each 10% increase in the proportion of time exposed to light (lux >0) resulted in a 0.67 hour decrease in RMSE. Similarly, a 1% increase in the proportion of time exposed to greater than 1000 lux reduced RMSE by 0.17 hours. Increasing mean illuminance by 100 lux reduces RMSE by 0.75 hours, and increasing median illuminance by 10 lux reduces RMSE by 0.53 hours. Increases of illuminance at the 95th percentile by 100 lux reduces RMSE by 0.08 hours. Day-to-day stability of light exposure patterns also improved accuracy. Improvements in accuracy were not strongly dependent on the daily pattern of the light (i.e., light fragmentation, PC2, PC3, PC4). The precision of the model, represented by entropy, was similarly impacted by the amount of light, though precision was also influenced by the pattern of the light. Whether the individual received greater light in the evening (higher PC2), a time at which phase delays are engendered by light exposure, or morning (lower PC2), a time at which phase advances are engendered by light exposure, did not significantly influence model performance.

Discussion

Our study evaluated the application of laboratory-based circadian models to real-world wearable light data, focusing on practical considerations such as initialization strategy, recording duration, and light exposure pattern in both regular schedule and shift worker populations. Using a probabilistic state estimation framework, we found that model performance improves with a greater number of initialization states but plateaus beyond ~48 initializations evenly-spaced in circadian phase. Longer recording durations (14 days)

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

improved accuracy in regular scheduled populations, but not in shift workers. In general, we found improved accuracy when individuals were exposed to higher illuminance light over longer durations with each 10% increase in the proportion of time exposed to light resulted in a 0.67 hour reduction in RMSE and increasing mean illuminance by 100 lux resulted in a reduction in RMSE by 0.75 hours. The pattern of this light exposure (i.e., fragmentation, PC3) appeared to influence the precision rather than the accuracy of model predictions.

These findings highlight the challenges of translating circadian models from controlled laboratory settings to real-world use with wearable sensors. In regular schedule participants, longer recordings and regular, structured, and bright light diets allow the models to converge to more accurate phase estimates, consistent with prior studies demonstrating more reliable performance under a regular light diet (Huang et al., 2021). In contrast, shift workers exhibited less bright and more irregular day-to-day light exposure patterns and showed minimal accuracy gains despite longer recordings. The decoupling between RMSE and entropy in shift workers suggests the model may converge on incorrect phase estimates, pointing toward a structural mismatch between model assumptions and input characteristics.

By accounting for uncertainty in the starting phase, our probabilistic initialization framework aligns with emerging evidence suggesting that errors in circadian phase estimation among shift workers are not primarily due to unknown initial conditions (Huang et al., 2021). The persistence of low accuracy despite this adjustment points to deeper limitations, either within the model structure, in our ability to accurately impute ocular light exposure from a wrist-worn device, or differences in light sensitivity among shift workers.

Accurate circadian phase estimation is not merely a convenience but a prerequisite for scalable field studies and timely interventions. Model-guided targeting has already been used to reduce the duration of DLMO assessments in shift workers from 24 hours to 5 hours and modeling results suggest that continued gains in estimator accuracy could shorten this window further (Lim et. al, 2025). Beyond laboratory logistics, wearable-derived markers of circadian timing and misalignment show robust associations with next-day mood and with depressive symptom burden in large, real-world cohorts, underscoring the value of precise, continuously updated phase for risk stratification and just-in-time care (Lee et al., 2024a). Parallel advances in digital psychiatry show that sleep- and circadian-derived features can prospectively predict depressive, manic, and hypomanic episodes in mood-disorder cohorts, underscoring actionable clinical use cases for phase-aware analytics (Lim et al., 2024). Finally, accurate model-informed circadian proxies also improve downstream inference (e.g., wearable sleep-stage classification), highlighting broader analytic gains from better phase estimation (Lee et al., 2024b).

Several limitations should be considered. Our data were drawn from two previously collected datasets in different geographic locations, introducing potential confounds such as seasonal variation, weather, device type, and compliance (i.e., obstruction by clothing) that covaried with the data type (regular schedule were collected in Northern California, the shift work were collected in Michigan). Additionally, model evaluation relied on a single in-laboratory phase marker. Evaluations that include multiple collections of both criterion standard phase markers with proximal light exposure patterns can be used to further

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

optimize these models by customizing individual sensitivities to light as well as other model parameters (Stone et al., 2020, Phillips et al., 2019).

Future work should prioritize two goals: identifying the source of model error and improving performance in irregular conditions. It remains unclear whether reduced accuracy in shift workers stems from data limitations (e.g., wrist vs ocular light exposure), irregular light structure, or underlying physiological changes associated with circadian disruption that would alter fundamental assumptions about how the clock responds to light. Our probabilistic framework could be applied across existing models to benchmark performance or used as a tool to personalize models via individual-specific parameter estimation. Prospective studies, collecting simultaneous ocular and wrist light exposure under rotating or irregular schedules with multiple in-lab circadian timing validation measures, would help disentangle data versus physiology. Moreover, advanced computational approaches, such as artificial intelligence, could help to disentangle and account for the origins of current uncertainty.

Conclusion

This work demonstrates the feasibility and limitations of applying laboratory-based circadian models to wearable light data in real-world settings. Using a probabilistic initialization framework, we identified conditions that support reliable phase estimation and revealed persistent failures in shift workers with irregular and low light exposure. These results underscore the importance of input structure over recording duration, challenge prior assumptions about initialization uncertainty, and point to a need for model adaptation and physiological validation in disrupted populations. Our findings lay the groundwork for more robust and scalable circadian tracking in free-living environments.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgments:

Research reported in this publication was supported by the NIH National Heart, Lung, and Blood Institute (F31HL170715), NIH National Institute of Child Health and Human Development (R01HD102344), and the Wu Tsai Human Performance Alliance at Stanford University. Data from night shift workers were obtained from a cohort recruited under NIH grant K23HL138166 (Principal Investigator: PC). The content is solely the responsibility of the authors and does not represent the official views of the National Institutes of Health or the Department of Veterans Affairs. Some of the computing for this project was performed on the Stanford University Sherlock cluster. We would like to thank Stanford University and the Stanford Research Computing Center for providing computational resources and support that contributed to these research results. The authors would also like to thank Renske Lok, Filipe Barata, and Zeitzer Circadian Research Lab members for their thoughtful feedback.

Funding Statement

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: Research reported in this publication was supported by the NIH National Heart, Lung, and Blood Institute (F31HL170715), NIH National Institute of Child Health and Human Development (R01HD102344), and the Wu Tsai Human Performance Alliance at Stanford University. Data from night shift workers were obtained from a cohort recruited under NIH grant K23HL138166 (Principal Investigator: PC).

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Data Availability:

Code supporting this work is available at https://github.com/Lara-Weed/circadian-wearable-init.

References

BONARIUS J, PAPATSIMPA C & LINNARTZ JP 2021. Parameter estimation in a model of the human circadian pacemaker using a particle filter. IEEE Trans Biomed Eng, 68, 1305–1316. [PubMed: 32970591]

BROWN LS, HILAIRE MA ST., MCHILL AW, PHILLIPS AJK, BARGER LK, SANO A, CZEISLER CA, DOYLE FJ III, & KLERMAN EB 2021. A classification approach to estimating human circadian phase under circadian alignment from actigraphy and photometry data. J Pineal Res, 71, e12745. [PubMed: 34050968]

BURGESS HJ, SAVIC N, SLETTEN T, ROACH G, GILBERT SS & DAWSON D 2003. The relationship between the dim light melatonin onset and sleep on a regular schedule in young healthy adults. Behav Sleep Med, 1, 102–14. [PubMed: 15600132]

CHENG P, WALCH O, HUANG Y, MAYER C, SAGONG C, CUAMATZI CASTELAN A, BURGESS HJ, ROTH T, FORGER DB & DRAKE CL 2021. Predicting circadian misalignment with wearable technology: validation of wrist-worn actigraphy and photometry in night shift workers. Sleep, 44, zsaa180. [PubMed: 32918087]

DUFFY JF & CZEISLER CA 2009. Effect of light on human circadian physiology. Sleep Med Clin, 4, 165–177. [PubMed: 20161220]

DUFFY JF & DIJK DJ 2002. Getting through to circadian oscillators: why use constant routines? J Biol Rhythms, 17, 4–13. [PubMed: 11837947]

FORGER DB, JEWETT ME & KRONAUER RE 1999. A simpler model of the human circadian pacemaker. J Biol Rhythms, 14, 532–7. [PubMed: 10643750]

HUANG Y, MAYER C, CHENG P, SIDDULA A, BURGESS HJ, DRAKE C, GOLDSTEIN C, WALCH O & FORGER DB 2021. Predicting circadian phase across populations: a comparison of mathematical models and wearable devices. Sleep, 44, zsab126. [PubMed: 34013347]

JOYCE DS, SPITSCHAN M, & ZEITZER JM 2022. Duration invariance and intensity dependence of the human circadian system phase shifting response to brief light flashes. Proc Biol Sci, 289(1970), 20211943 [PubMed: 35259981]

JOYCE DS, SPITSCHAN M, & ZEITZER JM 2022. Optimizing light flash sequence duration to shift human circadian phase. Biology (Basel), 11(12), 1807. [PubMed: 36552316]

KRONAUER RE, FORGER DB & JEWETT ME 1999. Quantifying human circadian pacemaker response to brief, extended, and repeated light stimuli over the photopic range. J Biol Rhythms, 14, 500–15. [PubMed: 10643747]

LEE MP, KIM DW, FANG Y, KIM R, BOHNERT ASB, SEN S, FORGER DB 2024. The real-world association between digital markers of circadian disruption and mental health risks. NPJ Digital Med, 7, 355.

LEE MP, KIM DW, MEYER C, WALCH O, FORGER DB 2024. The combination of topological data analysis and mathematical modeling improves sleep stage prediction from consumer-grade wearables. J Biol Rhythms, 39(6), 535–553. [PubMed: 39552521]

LIM D, JEONG J, SONG YM, CHO CH, YEOM JW, LEE T, LEE JB, LEE HJ, KIM JK 2024. Accurately predicting mood episodes in mood disorder patients using wearable sleep and circadian rhythm features. NPJ Digital Med, 7, 324.

LIM D, CHOI SJ, SONG YM, PARK HR, JOO EY, KIM JK 2025. Enhanced circadian phase tracking: a 5-h DLMO sampling protocol using wearable data. J Biol Rhythms, 40(3), 249–261. [PubMed: 40017128]

MAYER C, KIM DW, ZHANG M, LEE MP, FORGER DB, BURGESS HJ, MOON C 2025. Predicting circadian phase in community-dwelling later-life adults using actigraphy data. J Sleep Res, 34(4), e14425. [PubMed: 39647850]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

MOTT C, DUMONT G, BOIVIN DB & MOLLICONE D 2011. Model-based human circadian phase estimation using a particle filter. IEEE Trans Biomed Eng, 58, 1325–36. [PubMed: 21257371]

MROSOVSKY N 1999. Masking: history, definitions, and measurement. Chronobiol Int, 16, 415–29. [PubMed: 10442236]

NAJJAR RP, ZEITZER JM 2016. Temporal integration of light flashes by the human circadian system. J Clin Invest, 126(3), 938–47. [PubMed: 26854928]

PHILLIPS AJK, VIDAFAR P, BURNS AC, MCGLASHAN EM, ANDERSON C, RAJARATNAM SMW, LOCKLEY SW & CAIN SW 2019. High sensitivity and interindividual variability in the response of the human circadian system to evening light. Proc Natl Acad Sci U S A, 116, 12019–12024. [PubMed: 31138694]

HILAIRE MA ST, KLERMAN EB, KHALSA SB, WRIGHT KP JR., CZEISLER CA & KRONAUER RE 2007. Addition of a non-photic component to a light-based mathematical model of the human circadian pacemaker. J Theor Biol, 247, 583–99. [PubMed: 17531270]

HILAIRE MA ST, GRONFIER C, ZEITZER JM & KLERMAN EB 2007. A physiologically based mathematical model of melatonin including ocular light suppression and interactions with the circadian pacemaker. J Pineal Res. 43(3), 294–304. [PubMed: 17803528]

STONE JE, AUBERT XL, MAASS H, PHILLIPS AJK, MAGEE M, HOWARD ME, LOCKLEY SW, RAJARATNAM SMW, SLETTEN TL, 2019. Application of a limit-cycle oscillator model for prediction of circadian phase in rotating night shift workers. Sci Reports, 9, 11032.

STONE JE, POSTNOVA S, SLETTEN TL, RAJARATNAM SMW & PHILLIPS AJK 2020. Computational approaches for individual circadian phase prediction in field settings. Curr Opin Syst Biol, 22, 39–51.

WEED L, LOK R, CHAWRA D & ZEITZER J 2022. The impact of missing data and imputation methods on the analysis of 24-hour activity patterns. Clocks Sleep, 4, 497–507. [PubMed: 36278532]

WITTING W, KWA IH, EIKELENBOOM P, MIRMIRAN M & SWAAB DF 1990. Alterations in the circadian rest-activity rhythm in aging and Alzheimer's disease. Biol Psychiatry, 27, 563–72. [PubMed: 2322616]

WOELDERS T, BEERSMA DGM, GORDIJN MCM, HUT RA, WAMS EJ 2017. Daily light exposure patterns reveal phase and period of the human circadian clock. J Biol Rhythms, 32(3), 274–286. [PubMed: 28452285]

ZEITZER JM, DAVID R, FRIEDMAN L, MULIN E, GARCIA R, WANG J, YESAVAGE JA, ROBERT PH & SHANNON W 2013. Phenotyping apathy in individuals with Alzheimer disease using functional principal component analysis. Amer J Geriatric Psychiatr, 21, 391–397.

ZEITZER JM, DIJK DJ, KRONAUER R, BROWN E & CZEISLER C 2000. Sensitivity of the human circadian pacemaker to nocturnal light: melatonin phase resetting and suppression. J Physiol, 526, 695–702. [PubMed: 10922269]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

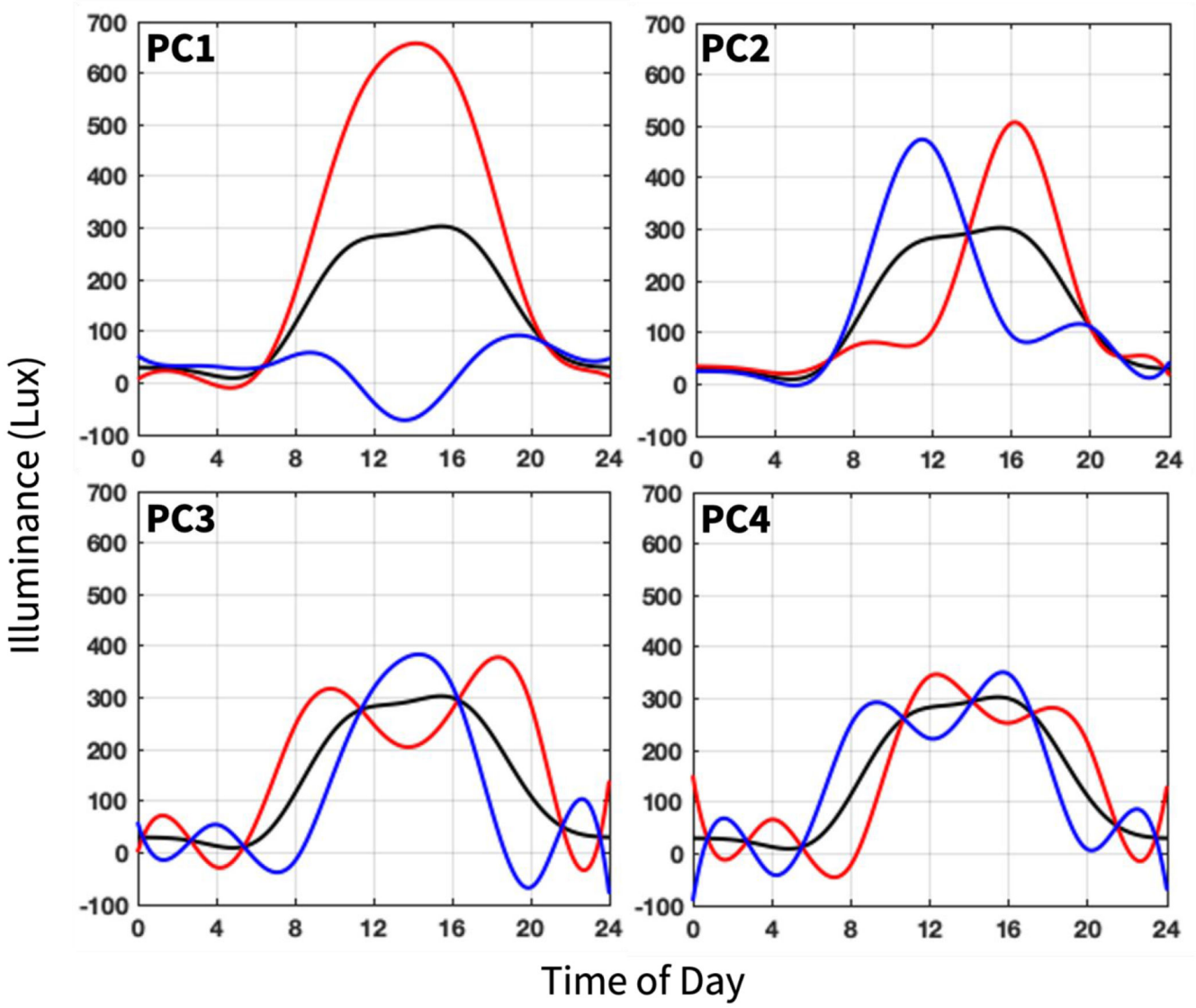


Figure 1:

Functional principal component analysis of light diets. Mean light exposure (black) is represented on all plots while standard deviations (red = positive, blue = negative) of principal components are ordered by percent variance of light diet explained. PC1 represents the magnitude of daylight exposure. PC2 represents morning and late afternoon light exposure dominance. PC3 represents the daytime duration of light exposure. PC4 represents preference toward earlier or later light exposure timing.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

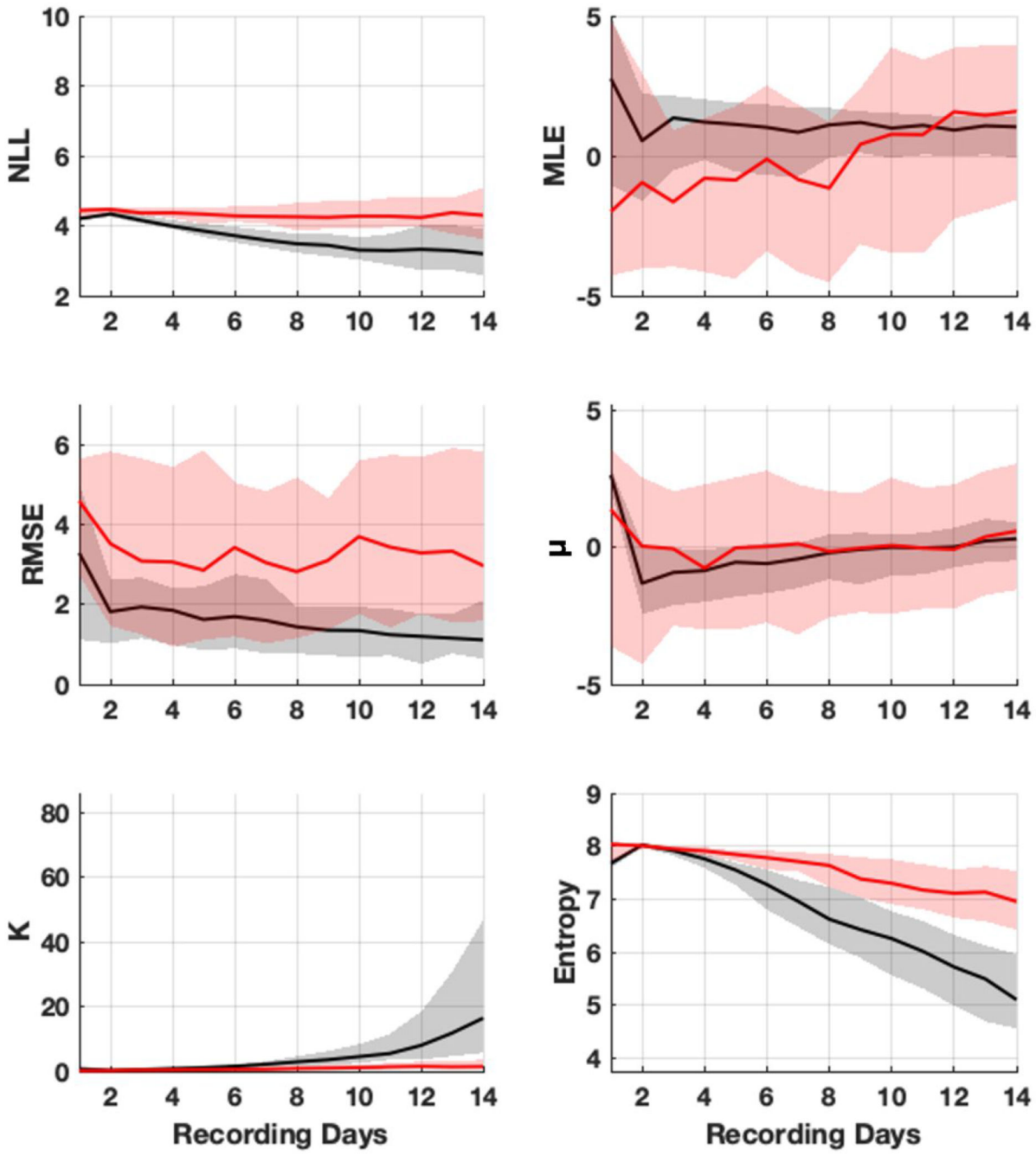


Figure 2:

Recording duration and model performance for the St. Hilaire et al. model in the regular schedule (black) and shift worker (red) datasets. Lines represent the median performance in predicting DLMO across participants and shading represents the IQR across participants for 1440 initializations evenly-spaced in phase. Longer recording durations resulted in improved model performance (lower RMSE and entropy) for the regular schedule dataset yielding a more precise (lower entropy, higher K) and accurate estimate (lower NLL and RMSE, closer to zero MLE and μ). The shift work dataset did not improve substantially with longer

recording durations. Recording lengths up to two weeks before the in-lab DLMO assessment are displayed.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

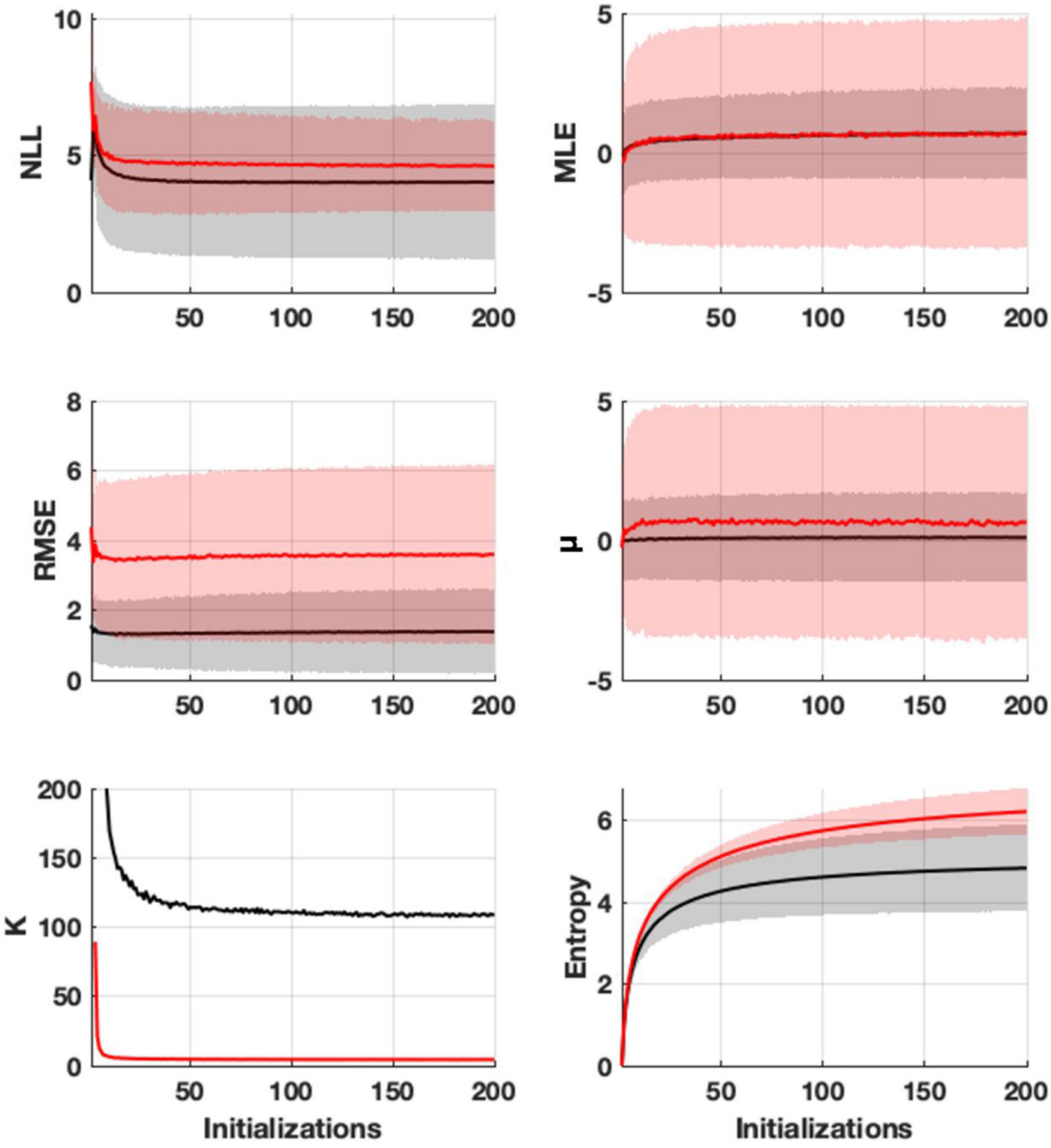


Figure 3:

Number of initializations and model performance for the St. Hilaire et al. model in the regular schedule (black) and shift worker (red) datasets. Lines represent the median across participants and shading represents the IQR across participants using recording lengths of 14 days. With too few initializations, model performance is mischaracterized (artificially low entropy and higher RMSE). Performance stabilized with more initializations for both regular schedule and shift workers. However, neither improved substantially with more initializations after stabilizing.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

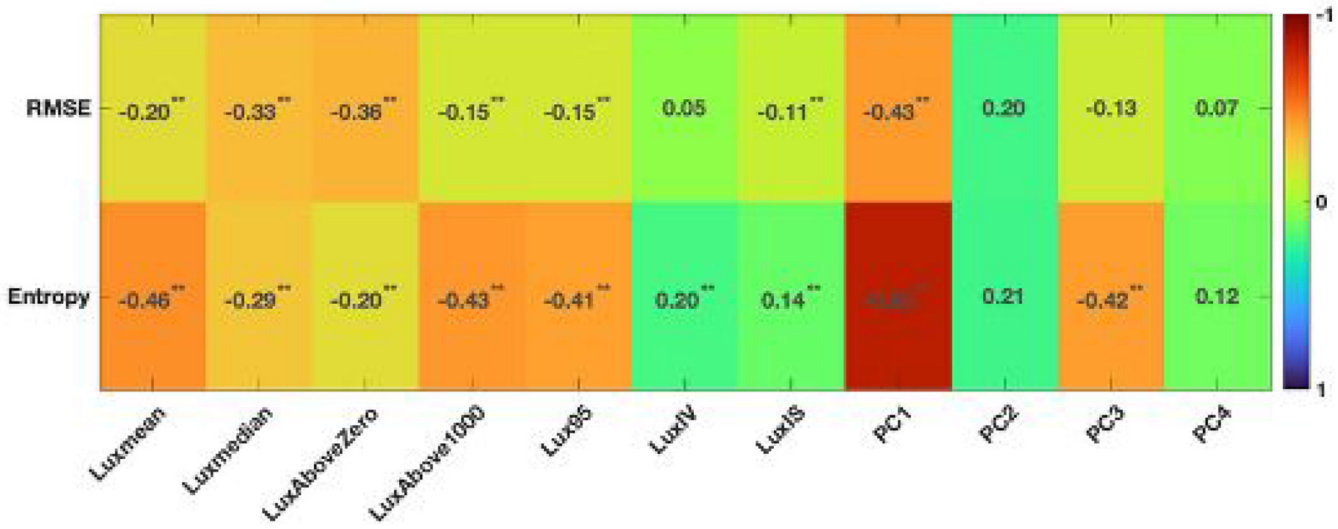


Figure 4:

Light diet feature and model performance correlations. Light brightness and duration measures (mean lux, median lux, percent of time above zero lux, percent of time greater than 1000 lux, and 95th percentile lux) show significant weak to moderate negative correlation (Pearson, $p<0.05$ (*), $p<0.01$ (**)) with RMSE and entropy, indicating that models increase in precision and accuracy. Regularity of light exposure patterns (IS) is weakly correlated with improved RMSE but higher entropy. Fragmentation of light exposure patterns (IV) is weakly correlated with higher entropy but shows no significant association with RMSE. The score of PC1 is moderately negatively correlated with RMSE, demonstrating improved accuracy with higher daytime light exposure amplitude.

Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

Table 1:

Characteristics of light recordings from wrist-worn wearable sensors in regular schedule and shift work datasets.

Metric	Regular Schedule N = 56, Median [IQR]	Shift Work N = 32, Median [IQR]	p-Value
Lux > 0, *Proportion*	0.69 [0.65-0.72]	0.49 [0.39-0.57]	<<0.01
Median Lux	14.59 [8.05-25.04]	0.04 [0.00-1.07]	<<0.01
Interdaily Stability	0.29 [0.24-0.39]	0.15 [0.08-0.21]	<<0.01
Recording length, *Days*	15.04 [14.08-16.07]	22.36 [14.36-33.91]	<0.01
Mean Lux	139.0 [97.20-207.50]	85.03 [66.08-137.36]	<0.01
Lux > 1000, *Proportion*	0.04 [0.02-0.06]	0.02 [0.02-0.04]	0.07
95th Percentile, *Lux*	464.4 [305.1-1476.2]	418.8 [309.0-857.7]	0.09
Intradaily Variability	0.94 [0.78-1.14]	0.91 [0.69-1.09]	0.27

Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

Table 2:

Functional principal component scores on light data from wrist-worn wearables in regular schedule and shift work datasets. PC1 represents the magnitude of daylight exposure. PC2 represents morning (more negative) and late afternoon (more positive) light exposure dominance. PC3 represents the daytime duration of light exposure. PC4 represents preference toward earlier (more negative) or later (more positive) light exposure timing. Mann-Whitney U tests were used to examine differences in temporal light exposure patterns between groups.

Functional Principal Component	Regular Schedule Median [IQR]	Shift Work Median [IQR]	p-Value
PC1	1000.0 [−2340.9 - 4761.5]	−4700.1 [−7516.2 - −943.80]	<<0.01
PC2	−366.38 [−1830.5 – 965.15]	1215.8 [−229.72 - 2173.4]	<0.01
PC3	−808.57 [−1724.9 - 1530.9]	−1272.1 [−1871.5 - 1026.6]	0.63
PC4	71.09 [−1089.4 - 583.87]	185.13 [−213.53 - 984.49]	0.26